

Chapter 5, Section 3

James Lee

September 7, 2026

1. Let X be a nonsingular projective variety of any dimension, let Y be a nonsingular subvariety, and let $\pi : \tilde{X} \rightarrow X$ be obtained by blowing up Y . Show that $p_a(\tilde{X}) = p_a(X)$.

Proof. The proof of (3.4) follows through for the more general case by (II, 8.24).

Proposition 0.1. *In the above setting, we have $\pi_* \mathcal{O}_{\tilde{X}} = \mathcal{O}_X$, and $R^i \pi_* \mathcal{O}_{\tilde{X}} = 0$ for $i > 0$. Therefore, $H^i(X, \mathcal{O}_X) \cong H^i(\tilde{X}, \mathcal{O}_{\tilde{X}})$ for all $i \geq 0$.*

□

2. Let C and D be curves on a surface X , meeting at a point P . Let $\pi : \tilde{X} \rightarrow X$ be the monoidal transformation with center P . Show that $\tilde{C} \cdot \tilde{D} = C \cdot D - \mu_P(C) \cdot \mu_P(D)$. Conclude that $C \cdot D = \sum \mu_P(C) \cdot \mu_P(D)$, where the sum is taken over all intersection points of C and D , including infinitely near intersection points.

Proof. Let E be the exceptional locus of $\pi : \tilde{X} \rightarrow X$. By (3.6), we have

$$\pi^* C = \tilde{C} + \mu_P(C)E.$$

Hence, we have

$$\tilde{C} \cdot \tilde{D} = (\pi^* C - \mu_P(C)E) \cdot (\pi^* D - \mu_P(D)E) = C \cdot D - \mu_P(C) \cdot \mu_P(D).$$

Assuming C and D are transversal, meeting at a finite number of points, the strict transforms of C and D under the blowing up of all intersection points will not meet, i.e., $\tilde{C} \cdot \tilde{D} = 0$. Each time an intersection point is blown up, $C \cdot D$ goes down by the product of the intersection multiplicities of each curve. □

3. Let $\pi : \tilde{X} \rightarrow X$ be a monoidal transformation, and let D be a very ample divisor on X . Show that $2\pi^* D - E$ is ample on $\tilde{X}$.

Proof.

□

5. Let $a_1, \dots, a_r$, $r \geq 5$, be distinct elements of k , and let C be the curve in $\mathbb{P}^2$ given by the (affine) equation $y^2 = \prod_{i=1}^r (x - a_i)$. Show that the point P at infinity on the y -axis is a singular point. Compute δ_P and $g(\tilde{Y})$, where $\tilde{Y}$ is the normalization of Y . Show in this way that one obtains hyperelliptic curves of genus $g \geq 2$.

7. For each of the following singularities at $(0, 0)$ in the plane, give an embedded resolution, compute δ_P , and decide which ones are equivalent.

- (a) $x^3 + y^5 = 0$
- (b) $x^3 + x^4 + y^5 = 0$
- (c) $x^3 + y^4 + y^5 = 0$
- (d) $x^3 + y^5 + y^6 = 0$
- (e) $x^3 + xy^3 + y^5 = 0$